

VIJAYAKUMAR

PYTHON BACKEND DEVELOPER

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PROFESSIONAL SUMMARY

Python Backend Developer with 4+ years of experience in developing web applications and AI-powered solutions across Financial Services, Investment Management, and Generative AI. Proficient in **Python, Django, Django Rest Framework, Flask, FastAPI, MySQL, PostgreSQL, Redis, Celery, AWS, Docker, and Git**, with hands-on experience in backend development, database design and optimization, cloud deployment, Generative AI applications, and **Retrieval-Augmented Generation (RAG) systems**.

TECHNICAL SKILLS

Programming Languages : Python, JavaScript, HTML, CSS
Frameworks & Libraries : Django, Django REST Framework, Flask, FastAPI, Django ORM
Databases & Caching : MySQL, PostgreSQL, SQLite, Redis, Celery
AI & Machine Learning : LangChain, Transformers, RAG
Cloud & DevOps : AWS, Docker, Git, GitHub
Software Practices : REST API, Agile Methodology, JWT Authentication & RBAC
Tools : VS Code, Postman

CORE COMPETENCIES

Python Backend Development, REST API Development, Django & FastAPI Development, Database Design & Optimization, AWS Cloud Services, Generative AI Applications, RAG Systems, Agile Development & Team Collaboration, Problem Solving & Debugging.

PROFESSIONAL EXPERIENCE

Python Backend Developer

Feb 2024 – Present

ADM

- Developed and maintained scalable web applications using Python, Django, Flask, FastAPI.
- Designed and implemented RESTful APIs and integrated third-party services.
- Built database-driven applications using MySQL and PostgreSQL.
- Implemented authentication, authorization, and role-based access control (RBAC).
- Collaborated with cross-functional teams in Agile/Scrum environments.
- Participated in code reviews, debugging, testing, deployment, and production support activities.
- Integrated AWS cloud services and supported application deployment and maintenance.

Wingsser Solutions

- Design, develop, and maintain scalable backend applications using Python, Django, FastAPI, and DRF.
- Designed and optimized MySQL databases for blockchain transactions, user accounts, content management, and messaging systems.
- Implemented authentication and authorization using JWT-based security mechanisms.
- Built features for content publishing, user engagement, direct messaging, wallet management, and digital asset tracking.
- Integrated blockchain-related functionalities including smart contract interactions and transaction management.
- Utilized Git for version control and participated in Agile/Scrum development processes.

PROJECTS

POC - Advanced Conversational Hybrid RAG System

Tools: Python, LangChain, Pinecone, FastEmbed, FastAPI

- Developed a Hybrid Retrieval-Augmented Generation (RAG) system using Python, FastAPI, LangChain, and Groq LLM for intelligent document question answering.
- Implemented document processing, including document loading, text chunking, embedding generation, and hybrid retrieval using BM25 and Pinecone Vector Database for accurate information retrieval.
- Built conversational memory, query rewriting, and REST APIs to support context-aware, multi-turn conversations while reducing hallucinations through retrieval-grounded responses.

Investment Research and Reporting Platform

Technologies: Django, Python, REST APIs , NumPy , Pandas ,PostgreSQL

- Developed a backend platform for investment research, portfolio analysis, risk assessment, and financial reporting for the Banking and Financial Services domain.
- Built REST APIs, integrated third-party financial data providers, and implemented secure communication and CORS-based frontend integration.
- Designed PostgreSQL database schemas and developed portfolio performance tracking and reporting features.

Decentralized Social & Content Sharing Platform

Technologies: Python, Django, Django REST Framework, MySQL, JWT, REST APIs, Git

- Developed RESTful APIs using Python, Django, and Django REST Framework for user authentication, user profiles, content publishing, direct messaging, and community engagement features.
- Designed and maintained MySQL database schemas for user accounts, posts, comments, direct messages, wallet transactions, and content management.
- Implemented JWT-based authentication and authorization, integrated backend APIs with frontend applications, and collaborated with cross-functional teams using Git in an Agile development environment

Blockchain Infrastructure Platform

Technologies: Python, Django, Django REST Framework, MySQL, REST APIs, Git

- Developed scalable backend services and RESTful APIs using Python, Django, and Django REST Framework for wallet management, blockchain transaction tracking, content management, and rewards distribution.
- Designed and optimized MySQL database schemas for user accounts, transaction history, digital assets, and platform metadata to ensure efficient data storage and retrieval.
- Implemented secure wallet management and smart contract integration while collaborating with cross-functional teams using Git in an Agile development environment.

POC - Automated Image Variant Generator

Technologies: Python, FastAPI, OpenCV, Pillow, PostgreSQL, Docker

- Developed an AI-powered application to generate multiple image variants from a single master image using image processing techniques.
- Built FastAPI REST APIs for image upload, variant generation, and metadata management with PostgreSQL.
- Integrated a pretrained DINOv2 model to generate image embeddings and measure similarity for consistent image variants.

EDUCATION

Master of Business Administration

Sakthi Institute of Information and Management Studies, Coimbatore | 7.1 CGPA

Bachelor of Computer Applications

Vidyasagar College of Arts and Science, Coimbatore | 7.8 CGPA